3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

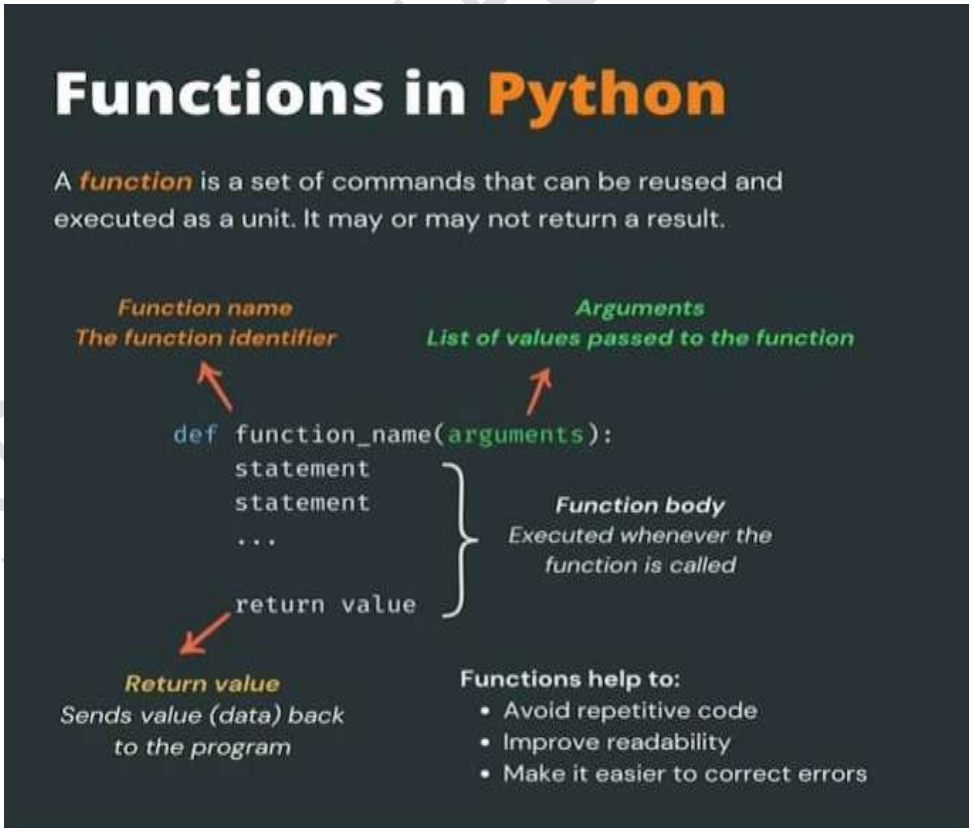
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 25: Functions in Python



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Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what a **Function** is.
- ✅ Learn why functions are useful.
- ✅ Create and call functions in Python.
- ✅ Use functions with and without parameters.
- ✅ Use functions that return values.
- ✅ Write simple programs using functions.

What is a Function?

A **Function** is a **reusable block of code** that performs a specific task.

Instead of writing the same code many times, we write it **once inside a function** and use it whenever needed.

Real-Life Example

Think of a **juice machine**.

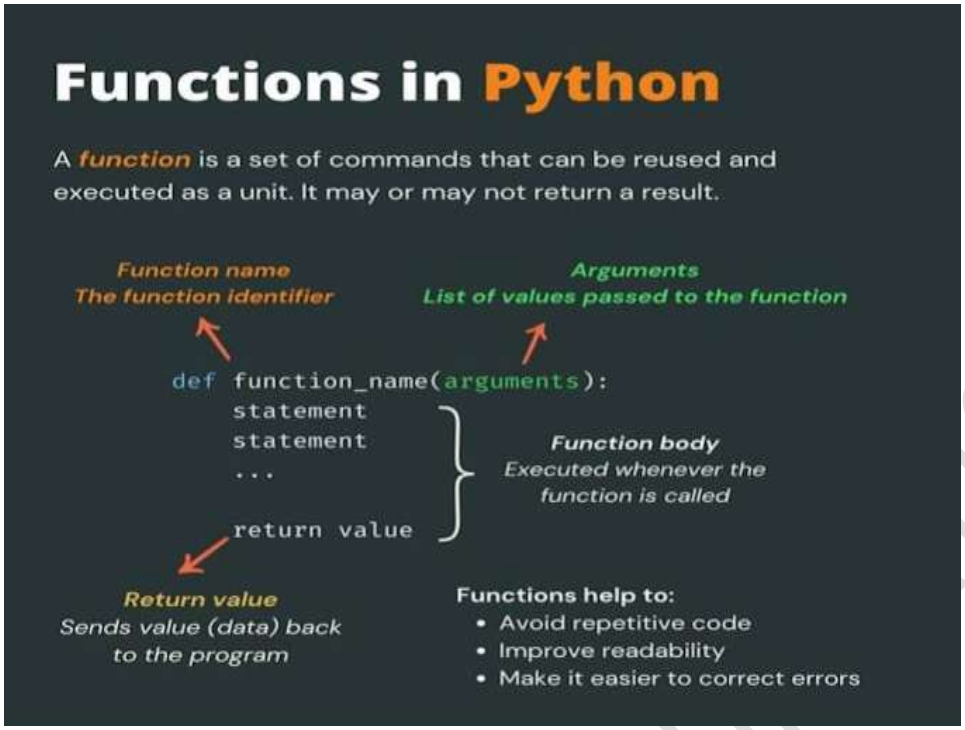
- 🍎 You put fruits into the machine.
- ⚙️ The machine processes them.
- 🥤 It gives you fresh juice.

You can use the same machine many times with different fruits.

A **function** works in the same way—it performs the same task whenever it is called.

Real-Life Example of a Function

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Why Do We Use Functions?

Functions help us:

- 🔁 Reuse code.
- 🧹 Keep programs short and organized.
- 📖 Make code easier to read.
- 🐞 Find and fix errors easily.
- 🤝 Share code with others.

Parts of a Function

A function has three main parts:

1. **Function Definition** (def)
2. **Function Call**

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3. **Function Body**

Function Syntax

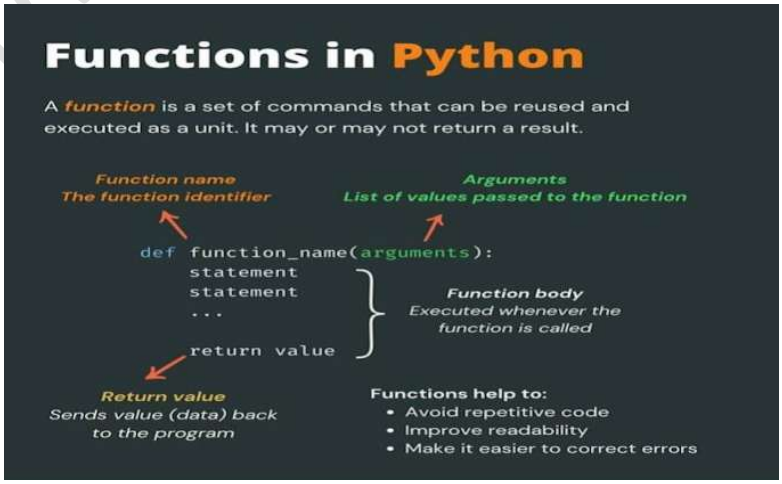
```
def function_name():
    # Code to execute

function_name()
```

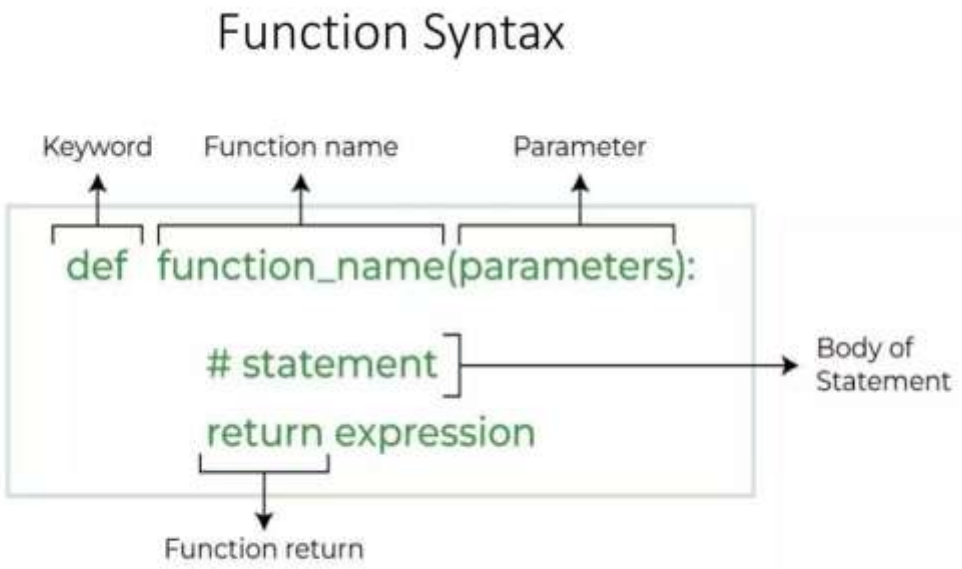
Understanding the Syntax

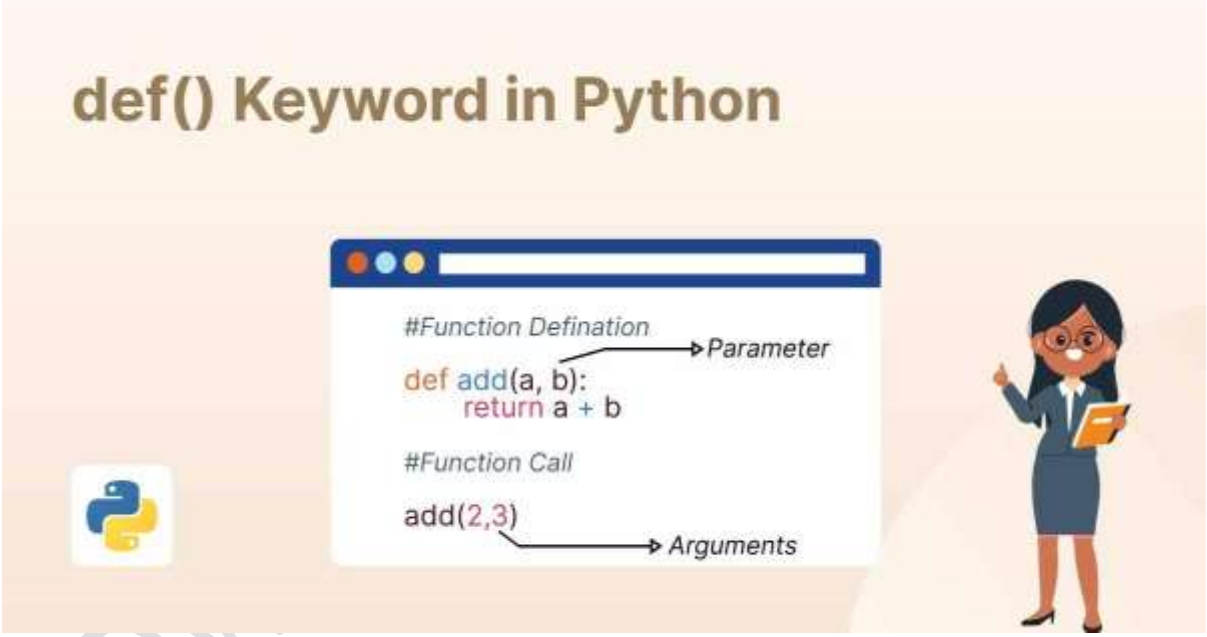
Keyword	Meaning
def	Defines a function
function_name	Name of the function
()	Parentheses
:	Starts the function block
Indentation	Code inside the function

Structure of a Function



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Example 1: Simple Function

```
def greet():
    print("Hello Students!")


greet()
```

Output

Hello Students!

Example 2: Function Called Multiple Times

```
def welcome():
    print("Welcome to Python")

welcome()
welcome()
welcome()
```

Output

Welcome to Python

Welcome to Python

Welcome to Python

Function with Parameters

A **parameter** is a value passed into a function.

Syntax

```
def function_name(parameter):
    # Code
```

Example 3

```
def greet(name):
    print("Hello", name)
```

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greet("Rahul")

Output

Hello Rahul

Example 4

```
def student(name, age):
    print("Name:", name)
    print("Age:", age)

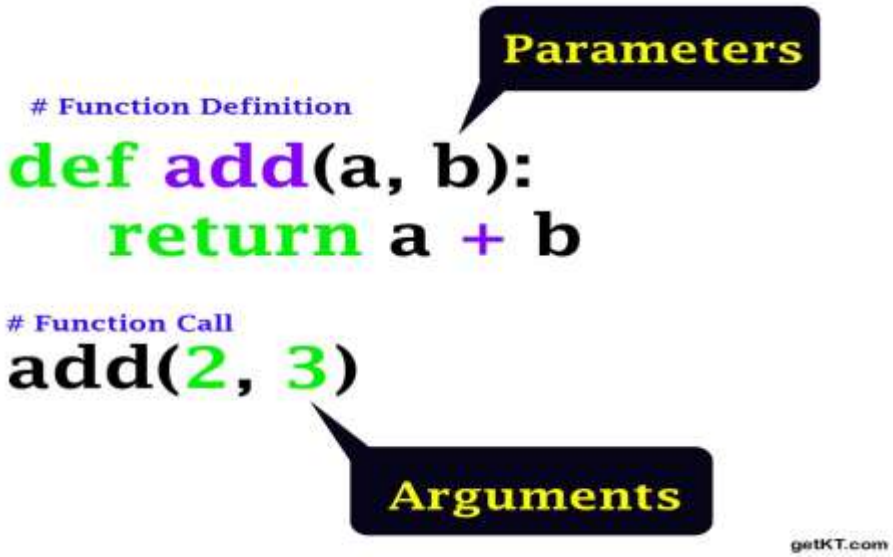
student("Asha", 13)
```

Output

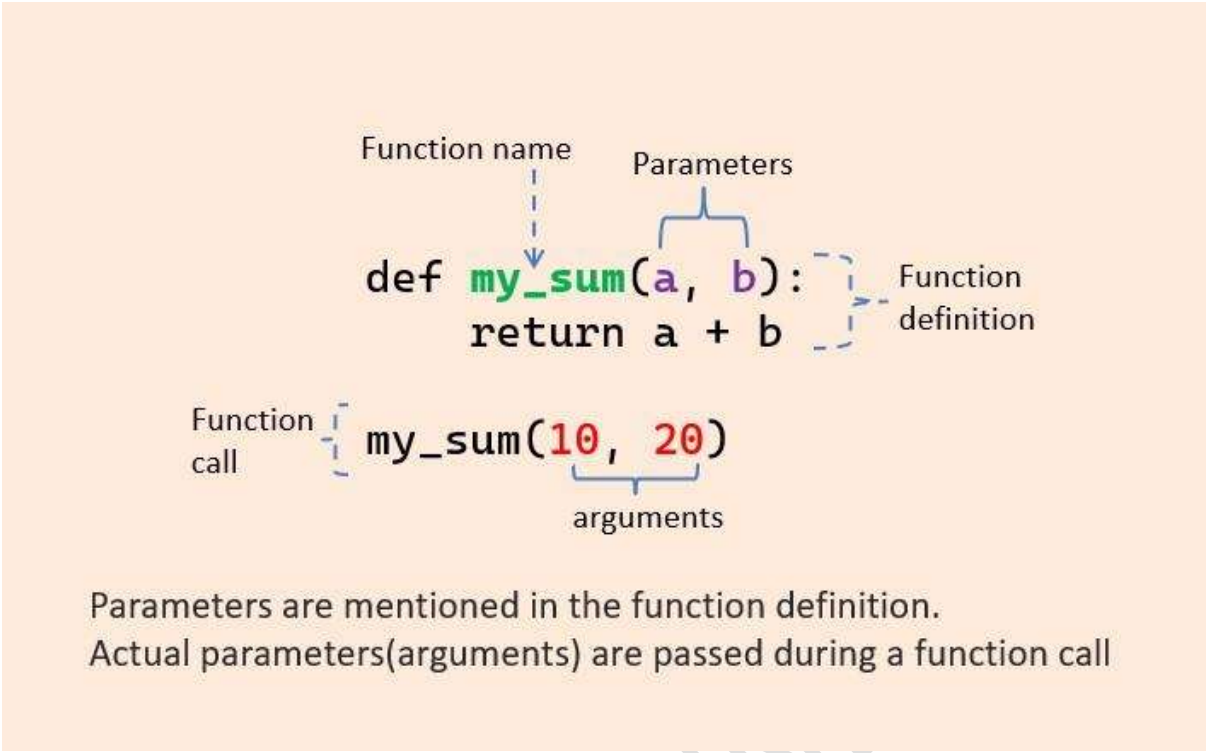
Name: Asha

Age: 13

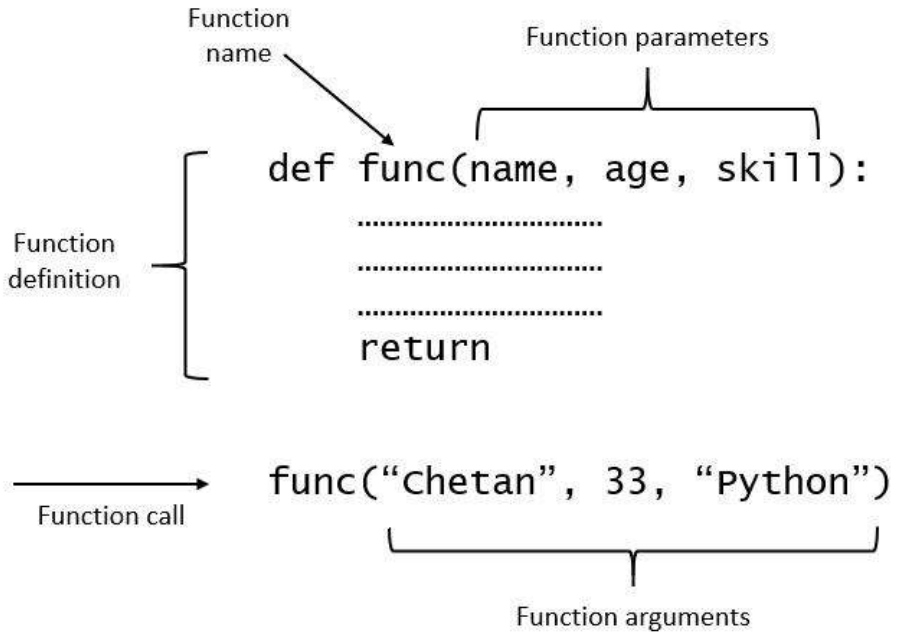
Function with Parameters



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Parameters are mentioned in the function definition.
Actual parameters(arguments) are passed during a function call



Function with Return Value

Sometimes a function **returns a result**.

We use the **return** keyword.

Example

```
def add(a, b):
    return a + b


result = add(10, 20)


print(result)
```

Output

30

Example 6

```
def square(number):
    return number * number


print(square(6))
```

Output

36

Example 7

```
def multiply(a, b):
    return a * b


answer = multiply(5, 8)
```

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```
print(answer)
```

Output

```
40
```

Built-in Functions

Python already provides many useful functions.

Function	Purpose
print()	Displays output
input()	Takes user input
len()	Finds the length
type()	Shows the data type
int()	Converts text to integer
str()	Converts value to string

Example

```
name = "Python"

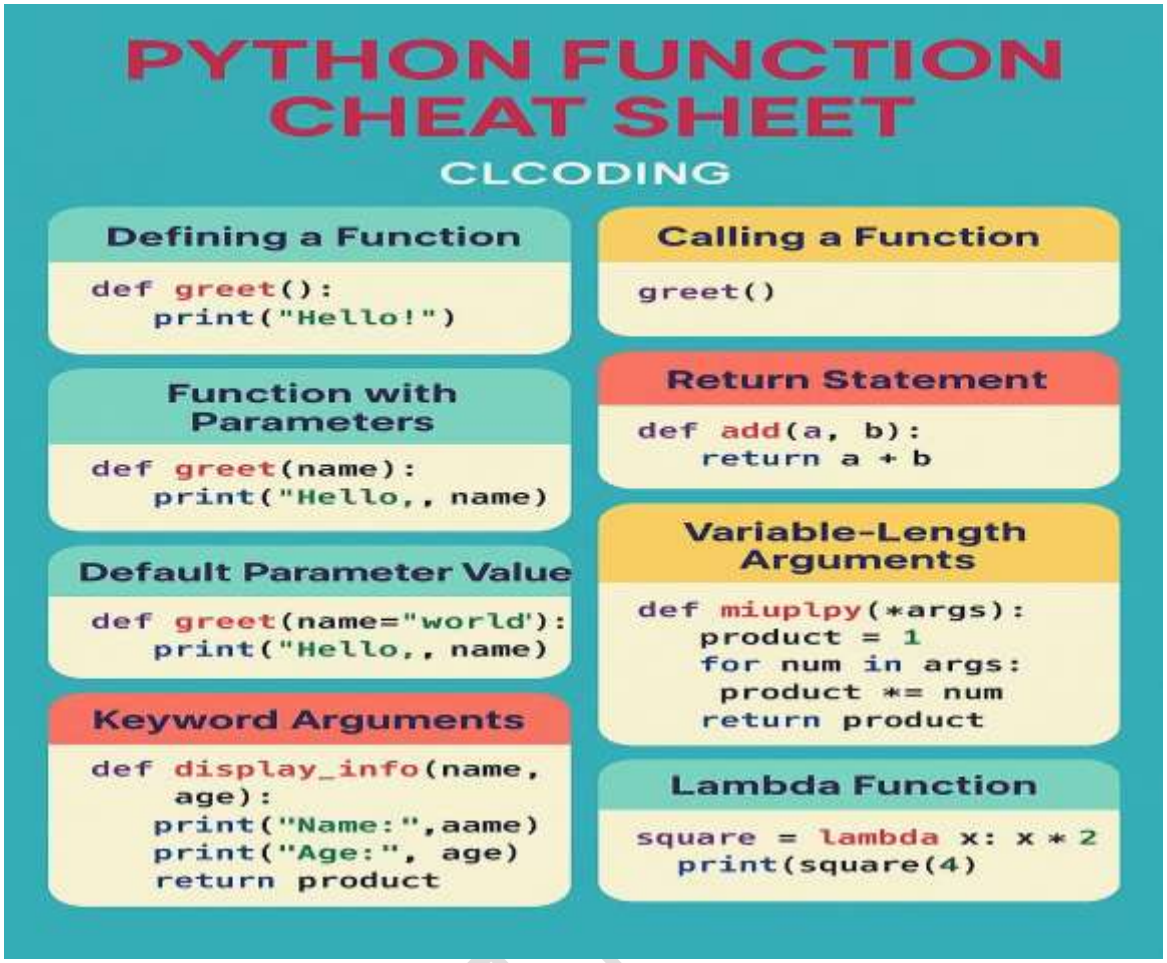
print(len(name))
```

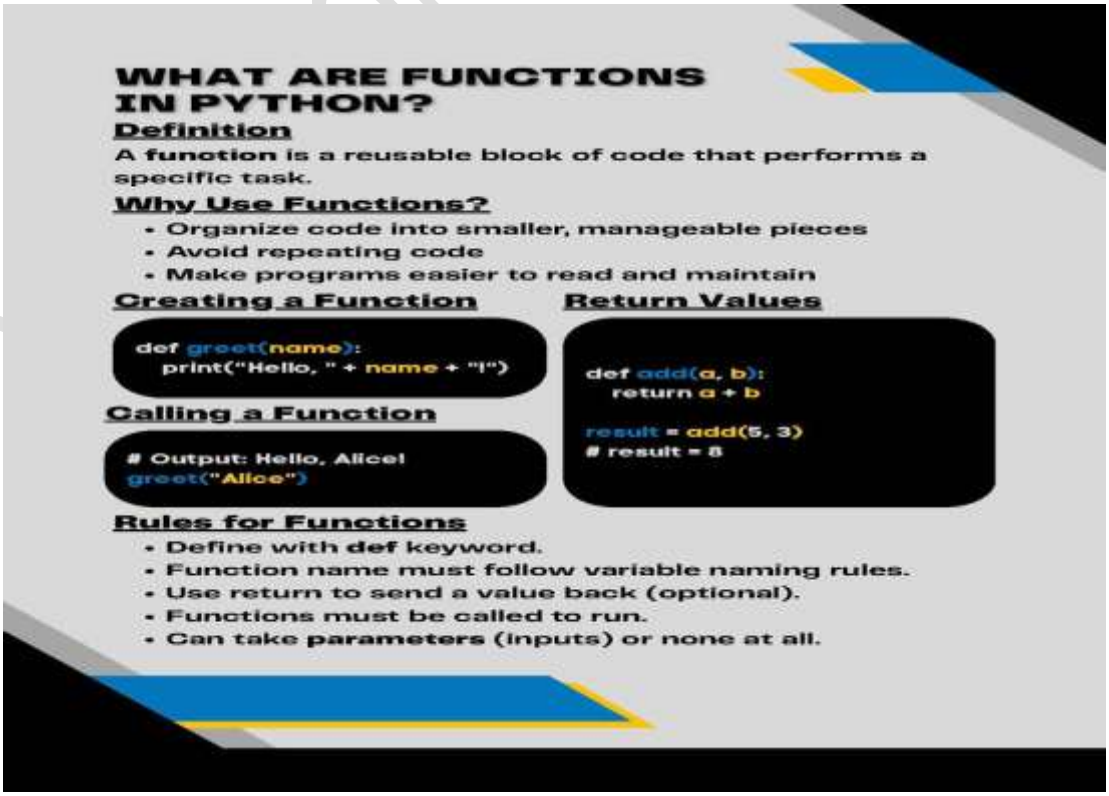
Output

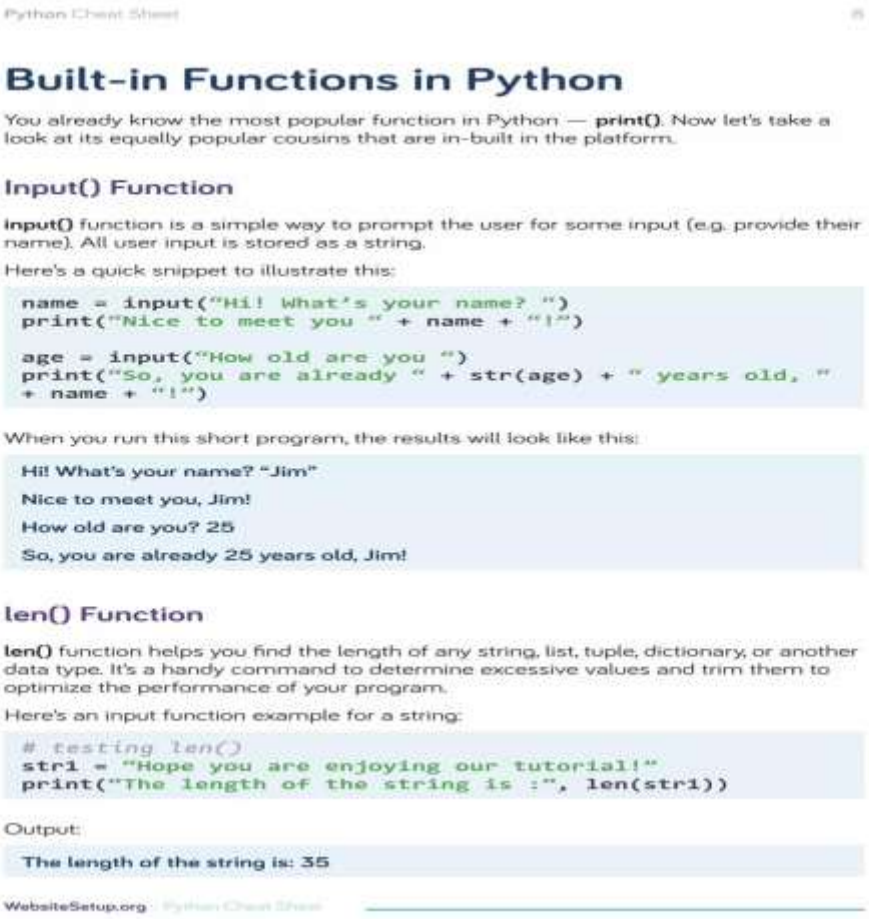
```
6
```

Built-in Functions

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PYTHON FUNCTION CHEAT SHEET
CLCODING
Defining a Function
def greet():
print("Hello!")
Calling a Function
greet()
Function with Parameters
def greet(name):
print("Hello,, name)
Return Statement
def add(a, b):
return a + b
Default Parameter Value
def greet(name="world'):
print("Hello,, name)
Variable-Length Arguments
def miuplpy(*args):
product = 1
for num in args:
product *= num
return product
Keyword Arguments
def display_info(name,
age):
print("Name:",aame)
print("Age:", age)
return product
Lambda Function
square = lambda x: x * 2
print(square(4)


WHAT ARE FUNCTIONS IN PYTHON?
Definition
A function is a reusable block of code that performs a specific task.
Why Use Functions?
• Organize code into smaller, manageable pieces
• Avoid repeating code
• Make programs easier to read and maintain
Creating a Function
def greet(name):
print("Hello, " + name + "!")
Return Values
def add(a, b):
return a + b
result = add(5, 3)
result = 8
Calling a Function
Output: Hello, Alice!
greet("Alice")
Rules for Functions
• Define with def keyword.
• Function name must follow variable naming rules.
• Use return to send a value back (optional).
• Functions must be called to run.
• Can take parameters (inputs) or none at all.



Python Cheat Sheet 6

Built-in Functions in Python

You already know the most popular function in Python — **print()**. Now let's take a look at its equally popular cousins that are in-built in the platform.

Input() Function

input() function is a simple way to prompt the user for some input (e.g. provide their name). All user input is stored as a string.

Here's a quick snippet to illustrate this:

```
name = input("Hi! What's your name? ")
print("Nice to meet you " + name + "!")

age = input("How old are you ")
print("So, you are already " + str(age) + " years old, "
+ name + "!")
```

When you run this short program, the results will look like this:

```
Hi! What's your name? "Jim"
Nice to meet you, Jim!
How old are you? 25
So, you are already 25 years old, Jim!
```

len() Function

len() function helps you find the length of any string, list, tuple, dictionary, or another data type. It's a handy command to determine excessive values and trim them to optimize the performance of your program.

Here's an input function example for a string:

```
# testing len()
str1 = "Hope you are enjoying our tutorial!"
print("The length of the string is :", len(str1))
```

Output:

```
The length of the string is: 35
```

WebsiteSetup.org - Python Cheat Sheet

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Practical Session

Practical Activity 1

Greeting Function

```
def greet():
    print("Good Morning!")


greet()
```

Practical Activity 2

Function with Name

```
def welcome(name):
    print("Welcome", name)

welcome("Rahul")
```

Practical Activity 3

Addition Function

```
def add(a, b):
    print("Sum =", a + b)

add(15, 25)
```

Practical Activity 4

Area of Rectangle

```
def area(length, breadth):
    print("Area =", length * breadth)

area(8, 5)
```

Practical Activity 5

Square Function

```
def square(number):
    return number * number
```

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```
print(square(9))
```

Practical Activity 6

Multiplication Function

```
def multiply(a, b):
    return a * b

print(multiply(7, 6))
```

Practical Activity 7

Student Information

```
def student(name, grade):
    print("Student:", name)
    print("Grade:", grade)

student("Anjali", "A")
```

Practical Activity 8

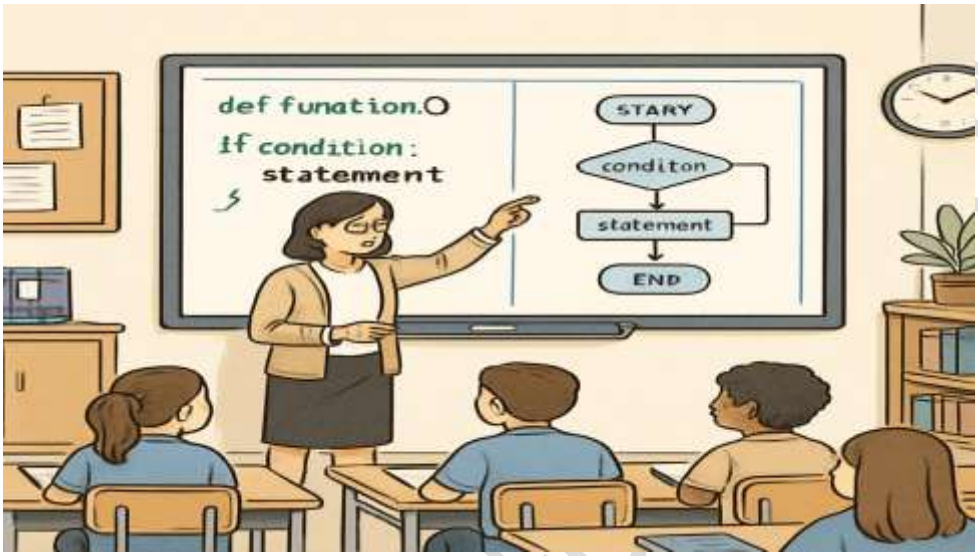
Check Even Number

```
def even(number):
    if number % 2 == 0:
        print("Even Number")
    else:
        print("Odd Number")
```

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even(12)

Classroom Coding Activity





Activity 1

Create a function to print your school name.

Activity 2

Create a function to add two numbers.

Activity 3

Create a function to display the multiplication table of a given number.

Activity 4

Create a function that returns the cube of a number.

Real-Life Applications

Situation	Function Used
Calculator	Addition, subtraction functions
ATM	PIN verification function
School Management	Student details function
Mobile App	Login function
Game	Score calculation function
Shopping Website	Bill calculation function

Common Mistakes

❌ **Forgetting Parentheses**

Wrong

```
def greet:
    print("Hello")
```

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✅ **Correct**

```
def greet():
    print("Hello")
```

❌ **Forgetting to Call the Function**

Wrong

```
def greet():
    print("Hello")
```

Nothing happens because the function is never called.

✅ **Correct**

```
def greet():
    print("Hello")

greet()
```

❌ **Incorrect Indentation**

Wrong

```
def greet():
print("Hello")
```

✅ **Correct**

```
def greet():
    print("Hello")
```

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Best Practices

- ✓ Give meaningful names to functions.
- ✓ Keep each function simple.
- ✓ Add comments to explain important functions.
- ✓ Reuse functions whenever possible.
- ✓ Test your function with different values.

Keywords

- **Function** – A reusable block of code.
- **def** – Keyword used to define a function.
- **Parameter** – Input given to a function.
- **Argument** – Actual value passed to a function.
- **Return** – Sends a result back from a function.
- **Built-in Function** – A function already provided by Python.

Lesson Summary

- A function is a reusable block of code.
- Functions are created using the **def** keyword.
- A function must be called to execute.
- Parameters allow functions to accept input values.
- The return statement sends results back to the program.
- Functions make programs shorter, cleaner, and easier to maintain.

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Quiz

Multiple Choice Questions

1. Which keyword is used to define a function?

A) if

B) while

C) def

D) for

✅ **Answer: C**

2. Which keyword returns a value from a function?

A) break

B) return

C) continue

D) print

✅ **Answer: B**

3. What is the output?

```
def hello():
    print("Hi")

hello()
```

A) Nothing

B) Hi

C) Error

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D) hello

✅ **Answer: B**

4. What is a parameter?

A) A loop

B) Input accepted by a function

C) A variable name

D) An operator

✅ **Answer: B**

5. Which of the following is a built-in Python function?

A) print()

B) school()

C) student()

D) myfunction()

✅ **Answer: A**

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Homework

1. Write a function to print your name.
2. Write a function to add two numbers.
3. Write a function to calculate the area of a rectangle.
4. Write a function that returns the square of a number.
5. Write a function to check whether a number is even or odd.

6. Write a function to display the multiplication table of a given number.
7. Create a function that prints your favorite subject.

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